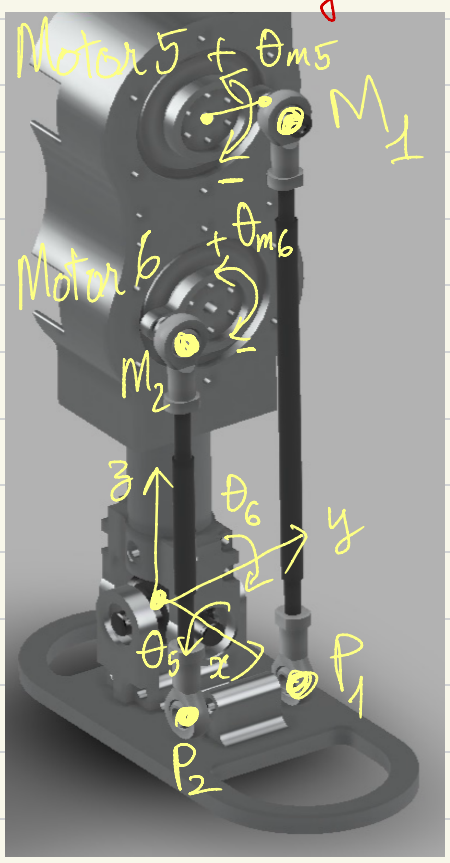
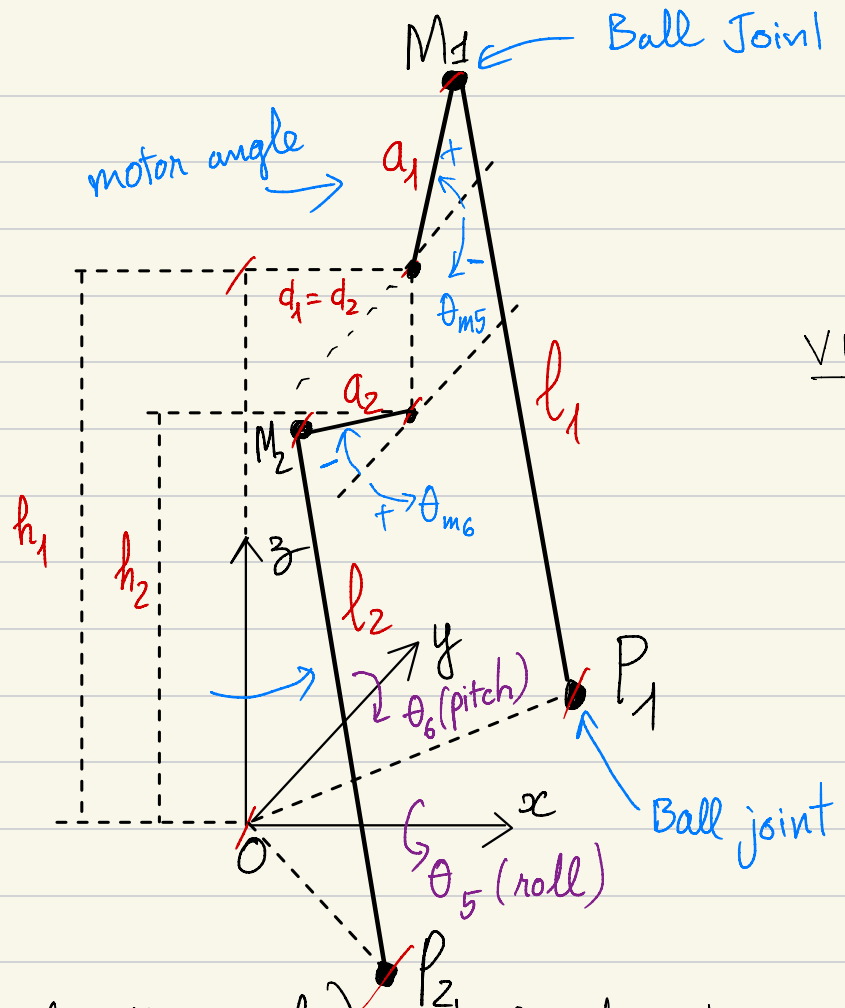
Left leg





Geometric / Design Params:

VREI: $a_1 = a_2 = 25.5\,mm$

$l_1 = 243.795\,mm$

$l_2 = 158.841\,mm$

$h_1 = 231\,mm$

$h_2 = 146\,mm$

$d_1 = d_2 = 42\,mm$

	P_{10}	P_{20}
x_0	[0.042	0.042] (m)
y_0	[0.0255	-0.0255] (m)
z_0	[-0.0222	-0.0222] (m)

Problem: Given θ_5 (roll angle) and θ_6 (pitch angle)

Find: θ_{m5} and θ_{m6}

Solution

As $\theta_5 = \theta_6 = 0 \Rightarrow \overrightarrow{OP_{10}} = r_{P10} = [P_{1x0} \;\; P_{1y0} \;\; P_{1z0}]^T$

const, represent the position of the ball joint on the foot

As we rotate the foot with angle θ_5 and θ_6, the position of P_1 is:

$$\overrightarrow{OP} = r_P = \underbrace{R_x(\theta_5)}_{\text{roll}}\,\underbrace{R_y(\theta_6)}_{\text{pitch}} \cdot r_{P10}$$

(1st rot: $R_x(\theta_5)$; 2nd rot: $R_y(\theta_6)$; intrinsic rotation)

$$= [x_{P_1} \;\; y_{P_1} \;\; z_{P_1}]^T$$

If roll then pitch → intrinsic rotation → post multiplication

If pitch then roll → extrinsic rotation → pre multiplication (2nd: $R_x(\theta_5)$, 1st: $R_y(\theta_6)$)

same result ← $R_x(\theta_5) \cdot R_y(\theta_6)$

Where rotation matrices:

$$R_y(\theta_6) = \begin{bmatrix} C_6 & 0 & S_6 \\ 0 & 1 & 0 \\ -S_6 & 0 & C_6 \end{bmatrix}$$

$$R_x(\theta_5) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C_5 & -S_5 \\ 0 & S_5 & C_5 \end{bmatrix}$$

$$R_x(\theta_5)\,R_y(\theta_6) = \begin{bmatrix} C_6 & 0 & S_6 \\ S_5S_6 & C_5 & -S_5C_6 \\ -C_5S_6 & S_5 & C_5C_6 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$$

$$= \begin{bmatrix} C_6x_0 + S_6z_0 \\ S_5S_6x_0 + C_5y_0 - S_5C_6z_0 \\ -C_5S_6x_0 + S_5y_0 + C_5C_6z_0 \end{bmatrix}$$

As motor 5 rotate an angle of θ_{m5}, the position of M (ball joint) is:

$$r_{M1} = \begin{bmatrix} d_1 \\ a_1 \cdot \cos\theta_{m5} \\ a_1 \sin\theta_{m5} + h_1 \end{bmatrix}, \quad r_{M2} = \begin{bmatrix} d_1 \\ -a_1 \cos\theta_{m6} \\ -a_1 \sin\theta_{m6} + h_1 \end{bmatrix}$$

As the length $\|PM\| = l_1 = const$, we have →

$$\|r_M - r_P\|^2 = l_1^2$$

$$\rightarrow (x_p - d)^2 + (y_p - a\,C_{m5})^2 + (z_p - aS_{m5} - h)^2 = l_1^2$$

Solve this to find θ_{m5}.

Now solve this eqn:

$$(x_{P_1} - d_1)^2 + (y_{P_1} - a_1 C_{m5})^2 + (z_{P_1} - a_1 S_{m5} - h_1)^2 = l_1^2$$

$$\Leftrightarrow \quad -2 y_{P_1} a_1 C_{m5} - 2 (z_{P_1} - h_1) a_1 S_{m5} + a_1^2 + (x_{P_1} - d_1)^2 + y_{P_1}^2 + (z_{P_1} - h_1)^2 = l_1^2$$

(u = $-2 y_{P_1} a_1$, v = $2 (z_{P_1} - h_1) a_1$)

$$\Leftrightarrow \quad \boxed{u C_{m5} + v S_{m5} = a_1^2 + (x_{P_1} - d_1)^2 + y_{P_1}^2 + (z_{P_1} - h_1)^2 - l_1^2}$$

(w = the right-hand side)

$$\Leftrightarrow \quad u C_{m5} + v S_{m5} = w \qquad (u, v, w \text{ are known})$$

Set $\alpha = \operatorname{atan2}(v, u)$; $R = \sqrt{u^2 + v^2}$

$$\begin{cases} u = R \cos\alpha \\ v = R \sin\alpha \end{cases} \quad \text{(polar coordinate)}$$

$$\hookrightarrow \quad R C_\alpha C_{m5} + R S_\alpha S_{m5} = w$$

$$\hookrightarrow \quad R C(\theta_{m5} - \alpha) = w$$

$$\rightarrow \quad \boxed{\theta_{m5} = \alpha \pm \arccos\left(\frac{w}{R}\right)}$$

For θ_{m6} : this becomes:

$$(-) u C_{m6} (-) v S_{m6} = a_1^2 + (x_p - d_1)^2 + y_{P_1}^2 + (z_{P_1} - h_1)^2 - l_1^2$$

This is why we us the vector sign_dir = [1 ; -1] in the code

(1 for θ_{m5}, -1 for θ_{m6})